University Ethics Form Version 3

Applicant Details (922)

Is this approval for a: (11240)

Type: (!/list-dropdown)

A2 - Student Project

What type of degree programme is being studied? (11319)

Type: (!/list-dropdown)

A2 - Postgraduate Taught (e.g. MSc / MA)

Name of Principal Researcher (11241)

Type: (S/text-short)

Vaishanth Bharadwaj

Please enter your email address (11258)

Type: (S/text-short)

v.bharadwaj2@ncl.ac.uk

Please select your school/academic unit (11242)

Type: (!/list-dropdown)

A3 - School of Engineering

Please enter the module code (11243)

Type: (S/text-short)

EEE8165

Please enter your supervisor's email (11259)

Type: (S/text-short)

patrick.degenaar@ncl.ac.uk

Please select your supervisor's school/unit: (11244)

Type: (!/list-dropdown)

A3 - School of Engineering

Project Details (923)

Project Title (11245)

Type: (S/text-short)

EEE8165 MSc Mock submission

Project Synopsis (11257)

Type: (T/text-long)

Your team will design, build and evaluate a wrist-worn embedded system that monitors a wearer's physical activity during everyday tasks. The device uses a microcontroller, motion sensor, rechargeable battery and wireless link to record and transmit data to a laptop or phone app. To evaluate the prototype, you will recruit fellow students to wear it during a short test session in which they carry out a set of everyday activities, such as sitting, walking and climbing stairs. You will analyse the recorded data to assess how accurately the system identifies each activity, then report your findings.

Project objectives

• Design and build a compact, wearable prototype using an embedded microcontroller and motion sensor.
• Develop firmware to acquire, process, and wirelessly transmit activity data.
• Plan and run a test session with student volunteers wearing the device.
• Collect, store and analyse the recorded data to evaluate classification accuracy.
• Report results, including system performance and limitations, in a technical report

Project start date (11260)

Type: (D/date)

01/01/2027

Project end date (11261)

Type: (D/date)

30/08/2027

Is the project externally funded? (11262)

Type: (!/list-dropdown)

A3 - No

Does your project involve collaborators outside of the University? (11265)

Type: (Y/yes-no)

No

Existing Ethics, Sponsorship & Responsibility (930)

Has ethical approval to cover this proposal already been obtained? (11267)

Type: (Y/yes-no)

No

Will anyone be acting as sponsor under the NHS Research Governance Framework for Health and Social Care? (11270)

Type: (Y/yes-no)

No

Do you have a Newcastle upon Tyne Hospitals (NUTH) reference? (11272)

Type: (Y/yes-no)

No

Will someone other than you (the principal investigator) or your supervisor (for student projects) be responsible for the conduct, management and design of the research? (11274)

Type: (Y/yes-no)

No

Animals (I) (924)

Please note that this question changed on 01/06/25 to capture activity with all animals

Does your proposed research involve the use of non-human animals i.e. the observation, capture or manipulation of animals OR their tissues (i.e. materials from animal such as skin, blood, bone, and other bodily fluids)?

(11246)

Type: (Y/yes-no)

No

NHS, Health & Social Care: Facilities, Staff & Patients (I) (925)

Will the study involve participants recruited by virtue of being NHS patients or service users, their dependents, their carers or human tissues or the use of NHS & Health/Social Care Facilities or otherwise require REC approval? (11247)

Type: (Y/yes-no)

No

Human Participants in a Non-Clinical Setting (I) (926)

Does the research involve human participants e.g. use of questionnaires, focus groups, observation, surveys or lab-based studies involving human participants? (11249)

Type: (Y/yes-no)

Yes

The default position is that projects involving human participants should be reviewed by a Faculty Ethics Committee. Exceptions are confined to studies in which interactions with participants are limited to those which most people would regard as trivial. For example, asking uncontentious questions in surveys or observing people undertaking normal activities. If in doubt please ask for advice.

Does the study involve any of the following?

(11250)

Type: (M/multiple-opt)

a. The study involves children or other vulnerable groups; including those who are relatively or absolutely incapable of protecting their own interests, or those in unequal relationships e.g. participants who are subordinate to the researcher(s) in a context outside the research? (11356)

b. The study requires the co-operation of a gatekeeper defined as someone who can exert undue influence) for initial access to the groups or individuals to be recruited e.g. students at school, members of a self-help group, or residents of a nursing home? NB. The IoN & School of Psychology volunteer pools are not considered gatekeepers in this case. (11357)

c. It is necessary for participants to take part in the study without their knowledge and consent e.g. covert observation of people in non-public places?.

(11358)

d. Deliberately misleading participants in any way? (11359)

e. Discussion of sensitive topics e.g. sexual activity or drug use?* (11360)

f. The administration of drugs, placebos or other substances (e.g. food substances, vitamins) to the study participants. (11361)

g. Invasive, intrusive or potentially harmful procedures of any kind? Including any procedure which is potentially harmful, has potential side effects or may reveal incidental findings. This includes, but is not limited to medical imaging procedures, magnetic stimulation procedures, hypnosis and vigorous exercise.* (11362)

h. Obtaining biological fluids (e.g. blood, saliva, urine), primary cells, or tissue samples?* (11363)

i. Pain or discomfort? (11364)

j. Psychological stress, anxiety, harm or negative consequences beyond that encountered in normal life? (11365)

k. Prolonged or repetitive testing i.e. more than 4 hours commitment or

attendance on more than two occasions? (11366)

l. Financial inducements (other than reasonable expenses and compensation for time)? (11379)

Data (I) (927)

Does the research involve the viewing, usage or transfer of sensitive data or personal data as defined by the General Data Protection Regulation (GDPR) or data governed by statute such as the Official Secrets Act 1989 / Terrorism Act 2006 , commercial contract or by convention e.g. client confidentiality? (If you are unsure please tick YES and complete the sub-questions). (11251)

Type: (Y/yes-no)

No

Environment and Responsible Research Innovation (I) (928)

Will the study potentially cause a direct or indirect negative impact on local communities, cultural heritage or the environment? (11253)

Type: (Y/yes-no)

No

International Projects (I) (929)

Will the research be conducted outside of the UK or European Economic Area (EEA) , or will it involve international collaborators outside the EEA? (11255)

Type: (Y/yes-no)

No

Next Steps (931)

Based on your responses your project has been categorised as (ethically) low risk and no further review is required before you start work. You will receive a formal approval email on submission of this form. Should your project change you may need to apply for new ethical approval. (11282)

Type: (X/boilerplate)

Supporting Documentation (940)

Please upload any documents (not uploaded elsewhere in the application) which you think are relevant to the consideration of your application. (11308)

Type: (|/upload-files)

Summary and Submission (941)

Thank you for completing the University's Ethical Review Form. Based on your answers the University is satisfied that your project has met its ethical expectations and grants its ethical approval.

Please be aware that if you make any significant changes to your project then you should complete this form again as further review may be required. Confirmation of this decision will be emailed to you.

Please complete the declaration to submit your application.

Declaration

I certify that:

(11314)

Type: (M/multiple-opt)

the information contained within this application is accurate. (11441)

the research will be undertaken in line with all appropriate, University, legal and local standards and regulations. (11442)

I have attempted to identify the risks that may arise in conducting this research and acknowledge my obligation to (and rights of) any participants. (11443)

no work will begin until all appropriate permissions are in place. (11444)

Projects with low risks. All. Confirmation email.

(11338)

Type: (S/text-short)

Based on your answers, the University Ethics Committee grants its approval for you to start working on your project. Please be aware that if you make any significant changes to your proposal then you should complete this form again, as further review may be required. This confirmation may be used within a research portfolio as evidence of ethical approval. Please note: this confirmation will be the only correspondence you should expect to receive as evidence of ethical approval. There will be no other confirmation provided. You may now proceed with research. If you have any queries, please review the internal and external ethics FAQ pages before contacting res.policy@ncl.ac.uk.